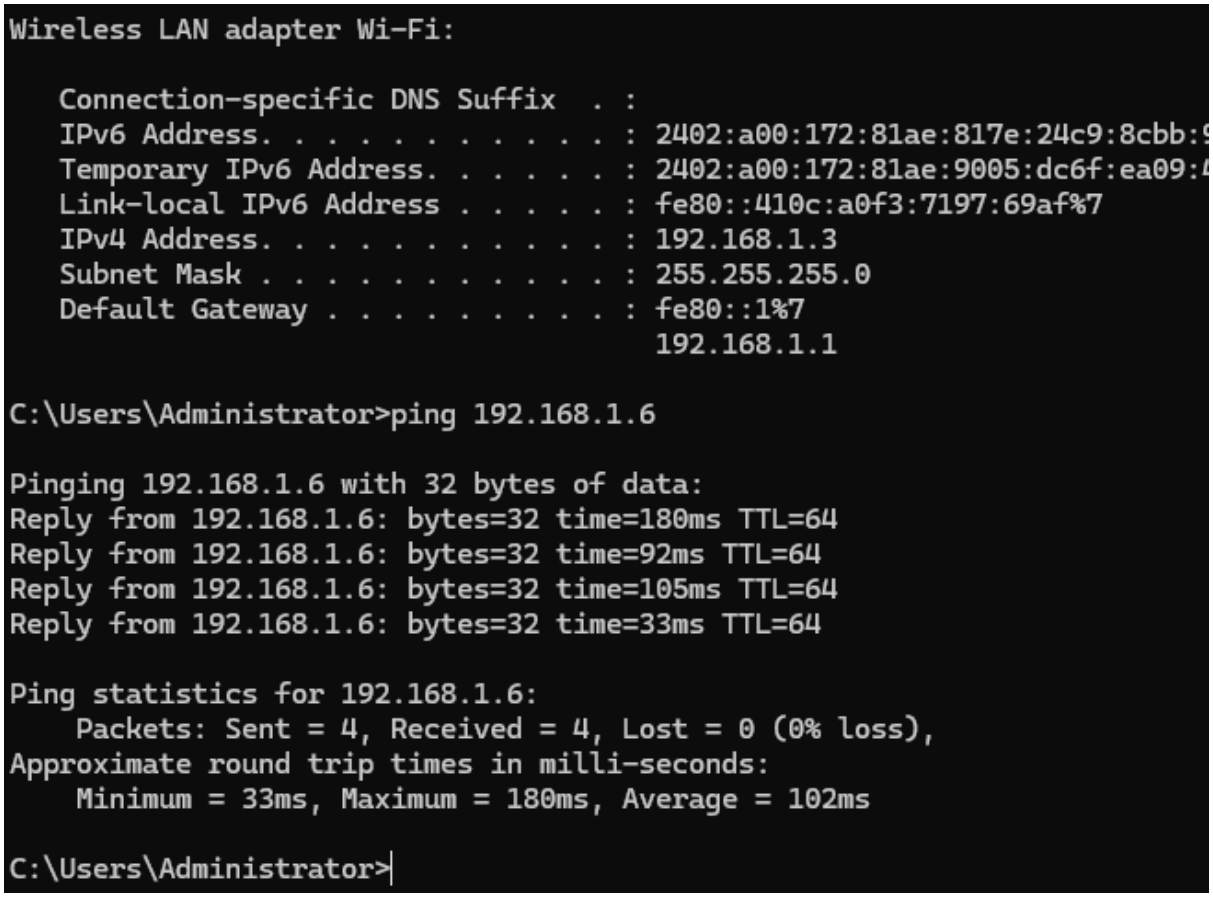
Day 5: Network Testing Practical (1 hour): Connect devices to a network and test connectivity (ping, traceroute). Task: Write a report on how to set up and test a basic network.

TASK- 1

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

➔



IP v 4 is my pc IP address

After ping my phone IP address

Both are connected to same network

And ping 192.168.1.6 (phone) is responding so all connections are good

TASK – 2

Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

➔

```
C:\Users\Administrator>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482a:400::]
over a maximum of 30 hops:

  1     *        *        *     Request timed out.
  2     4 ms     3 ms     3 ms  undefined.hostname.localhost [2402:a00:172:1::1111]
  3     4 ms     3 ms     3 ms  undefined.hostname.localhost [2402:a00:60::1005]
  4    27 ms    27 ms    26 ms  undefined.hostname.localhost [2402:a00:60::36]
  5    27 ms    27 ms    27 ms  2001:4860:482a:400::

Trace complete.
```

Step 1: Open Command Prompt on your Windows laptop.

Step 2: Type this command and press Enter:

`tracert www.youtube.com`

Step 3: Wait for the command to finish. It may take a few minutes.

Step 4: Count the hops. Look at the numbers at the beginning of each output line. Each number represents a hop along the route.

Objective: To identify the number of network hops required to reach YouTube.

Command used: `tracert www.youtube.com`

Procedure: I opened Command Prompt on my Windows laptop, entered the traceroute command, and waited for the results. I observed the hop numbers and response times displayed in the output.

Result: Number of hops observed: 5

Conclusion: The tracert command helped me observe the network route toward YouTube and the response times of the hops that replied.

A hop is one step that data takes through a network device, usually a router, when travelling from your laptop to a destination like YouTube.

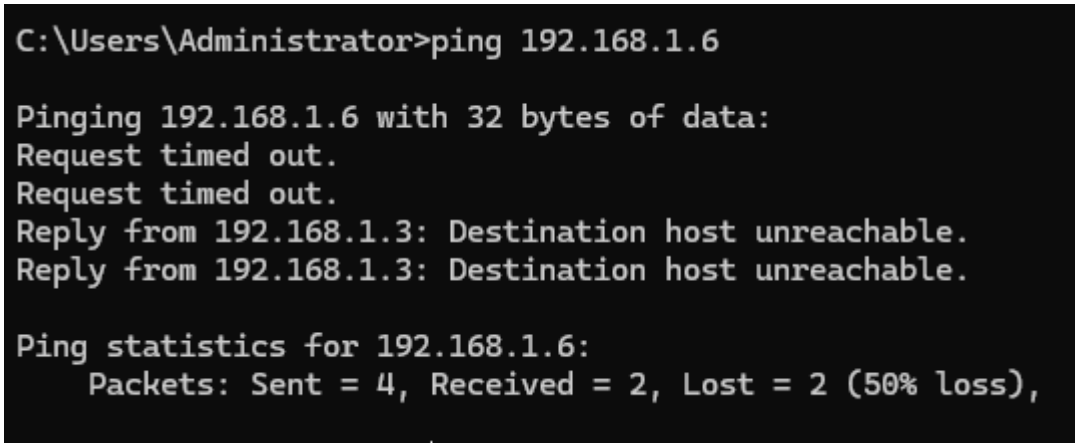
TASK - 3

Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

➔

Steps to perform:

1. Connect your laptop and your friend's phone to the same Wi-Fi.
2. Find your friend's phone IP address.
3. Disconnect your friend's phone from the Wi-Fi.
4. On your laptop, open Command Prompt (`cmd`).
5. Type the following command, replacing the example IP with your friend's phone IP:

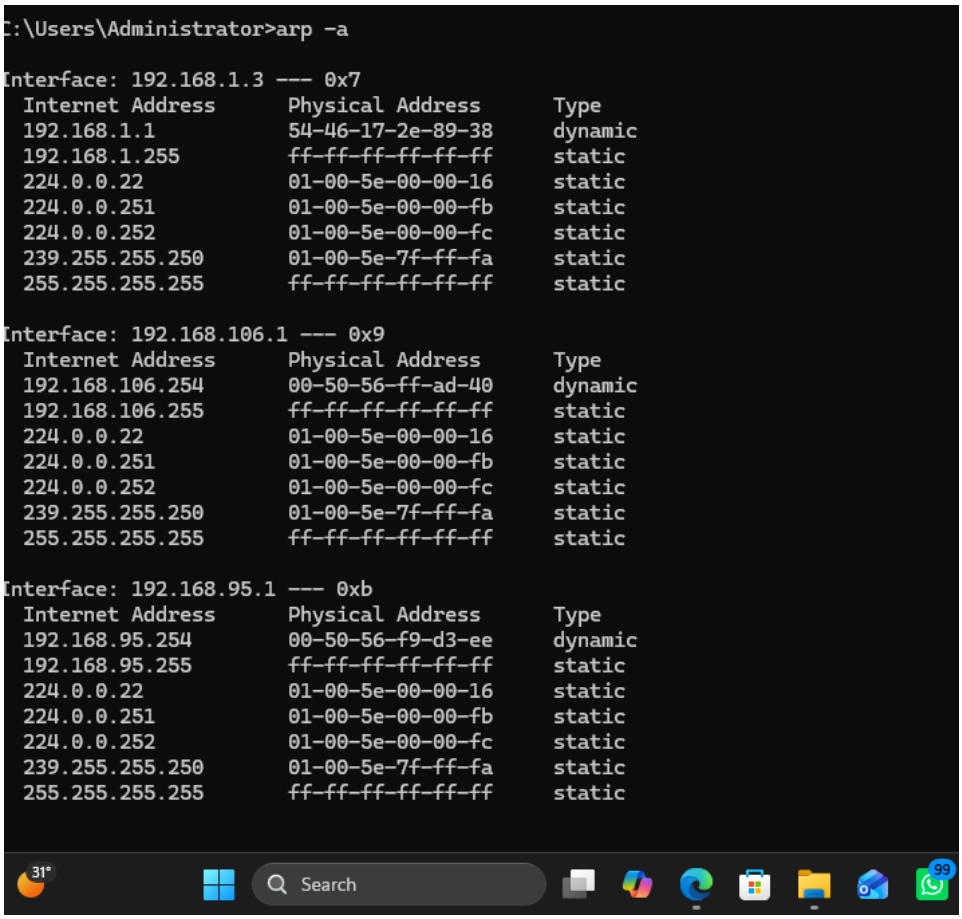


After disconnecting my phone wifi internet

TASK – 4

Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

➔



The `arp -a` command displays IP addresses and their associated physical (MAC) addresses that my laptop has learned on the local network. I used this information to help identify devices on my home Wi-Fi.

`arp -a` may not show every connected device. It also does not directly tell you the device owner's name. You may need to check your router's connected-device list or compare each device's IP address in its Wi-Fi settings.

No.	IP Address	MAC Address	Possible device
1	`192.168.1.1`	`54-46-17-2e-89-38`	Router or gateway
2	`192.168.106.254`	`00-50-56-ff-ad-40`	Possible virtual network gateway
3	`192.168.95.254`	`00-50-56-f9-d3-ee`	Possible virtual network gateway

TASKS -5

Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

➔

Day 5: Network Testing Practical Report

Objective: To set up a local network, test connectivity using the `ping` command, trace the route to a website using `tracert`, and identify devices using `arp -a`.

1. Network Setup

First, I connected my laptop and my friend's mobile phone to the same Wi-Fi network. I opened Command Prompt on my Windows laptop and used the `ipconfig` command to check the network configuration and IP address.

2. Connectivity Test Using Ping

I used the `ping` command to test communication between my laptop and my friend's phone.

Command used: `ping [Phone IP Address]`

When both devices were connected to Wi-Fi, I checked whether the phone responded to the ping request. Next, I disconnected the phone from Wi-Fi and tried pinging it again. The connection test could show a message such as “Request timed out” if the laptop did not receive a response.

3. Traceroute Test

I used the following command to trace the network route to YouTube.

Command used: `tracert www.youtube.com`

Result: The destination was reached in 5 hops. The first hop showed “Request timed out,” while hops 2, 3, 4, and 5 responded successfully. The final hop reached the destination.

4. Identifying Devices Using ARP

I ran the following command to view IP and MAC address information learned by my laptop.

Command used: `arp -a`

I used the output to prepare a table of IP addresses and their possible devices, such as the router, laptop, and mobile phone. Some devices could not be identified directly from the output.

5. Conclusion

Through this practical, I learned how to connect devices to the same Wi-Fi network, test connectivity using `ping`, trace a route using `tracert`, and view local network address information using `arp -a`. I also learned that a timeout does not always mean a device is disconnected, because a device or network may block ping requests.

